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INSTITUTE OF TECHNOLOGY

DHULE (M.S.)

DEPARMENT OF COMPUTER ENGINEERING

Remark

Subject : Machine Learning Lab

Name:- **Roll No. :**

Class: T.Y Comp Engg. **Batch :** T **Division:** _______

Expt. No. : 03 **Performance Date :** / /

Signature

Title : Write a program to implement the Implement Simple Linear Regression and Multiple Linear Regression for a sample training data set stored as a .CSV file. Compute the accuracy of the classifier, considering few test data sets and Visualize model with Python.

Rubrics:

Date of Submission:		
Marks Split up to	**Maximum Marks**	**Marks Obtained**
Performance/conduction		
Report Writing		
Attendance		
Viva/Oral		
Total Marks		

Aim :

The main aim of this lab is to demonstrate the how to implement the simple linear regression and multiple linear regression Python.

Secondary aim of this lab is to undertsand

- What linear regression is
- What linear regression is used for
- How linear regression works
- How to implement linear regression in Python, step by step

Software Required : Jupyter Notebook , Python 3.6, Or Google Colab

Dataset: Salary_Data.csv, 50_Startups.csv

Theory :	

Regression

Regression analysis is one of the most important fields in statistics and machine learning. There are many regression methods available. Linear regression is one of them.

What Is Regression?

Regression searches for relationships among variables.

For example, you can observe several employees of some company and try to understand how their salaries depend on the **features**, such as experience, level of education, role, city they work in, and so on.

This is a regression problem where data related to each employee represent one **observation**. The presumption is that the experience, education, role, and city are the independent features, while the salary depends on them.

Similarly, you can try to establish a mathematical dependence of the prices of houses on their areas, numbers of bedrooms, distances to the city center, and so on.

Generally, in regression analysis, you usually consider some phenomenon of interest and have a number of observations. Each observation has two or more features. Following the assumption that (at least) one of the features depends on the others, you try to establish a relation among them.

In other words, **you need to find a function that maps some features or variables to others sufficiently well**.

The dependent features are called the **dependent variables**, **outputs**, or **responses**.

The independent features are called the **independent variables**, **inputs**, or **predictors**.

Regression problems usually have one continuous and unbounded dependent variable. The inputs, however, can be continuous, discrete, or even categorical data such as gender, nationality, brand, and so on.

It is a common practice to denote the outputs with y and inputs with x. If there are two or more independent variables, they can be represented as the vector $\mathbf{x} = (x_1, \ldots, x_r)$, where r is the number of inputs.

When Do You Need Regression?

Typically, you need regression to answer whether and how some phenomenon influences the other or **how several variables are related**. For example, you can use it to determine *if* and *to what extent* the experience or gender impact salaries.

Regression is also useful when you want **to forecast a response** using a new set of predictors. For example, you could try to predict electricity consumption of a household for the next hour given the outdoor temperature, time of day, and number of residents in that household.

Regression is used in many different fields: economy, computer science, social sciences, and so on. Its importance rises every day with the availability of large amounts of data and increased awareness of the practical value of data.

Linear Regression

Linear regression is probably one of the most important and widely used regression techniques. It's among the simplest regression methods. One of its main advantages is the ease of interpreting results.

Problem Formulation

When implementing linear regression of some dependent variable y on the set of independent variables $\mathbf{x} = (x_1, \ldots, x_r)$, where r is the number of predictors, you assume a linear relationship between y and $\mathbf{x}$: $y = \beta_0 + \beta_1 x_1 + \cdots + \beta_r x_r + \varepsilon$. This equation is the **regression equation**. $\beta_0, \beta_1, \ldots, \beta_r$ are the **regression coefficients**, and ε is the **random error**.

Linear regression calculates the **estimators** of the regression coefficients or simply the **predicted weights**, denoted with $b_0, b_1, \ldots, b_r$. They define the **estimated regression function** $(\mathbf{x}) = b_0 + b_1 x_1 + \cdots + b_r x_r$. This function should capture the dependencies between the inputs and output sufficiently well.

The **estimated** or **predicted response**, $(\mathbf{x}_i)$, for each observation $i = 1, \ldots, n$, should be as close as possible to the corresponding **actual response** y_i. The differences y_i - $(\mathbf{x}_i)$ for all observations $i = 1, \ldots, n$, are called the **residuals**. Regression is about determining the **best predicted weights**, that is the weights corresponding to the smallest residuals.

To get the best weights, you usually **minimize the sum of squared residuals** (SSR) for all observations $i = 1, \ldots, n$: SSR = $\Sigma_i(y_i - f(\mathbf{x}_i))^2$. This approach is called the **method of ordinary least squares**.

Regression Performance

The variation of actual responses y_i, $i = 1, \ldots, n$, occurs partly due to the dependence on the predictors $\mathbf{x}_i$. However, there is also an additional inherent variance of the output.

The **coefficient of determination**, denoted as R^2, tells you which amount of variation in y can be explained by the dependence on $\mathbf{x}$ using the particular regression model. Larger R^2 indicates a better fit and means that the model can better explain the variation of the output with different inputs.

The value $R^2 = 1$ corresponds to SSR = 0, that is to the **perfect fit** since the values of predicted and actual responses fit completely to each other.

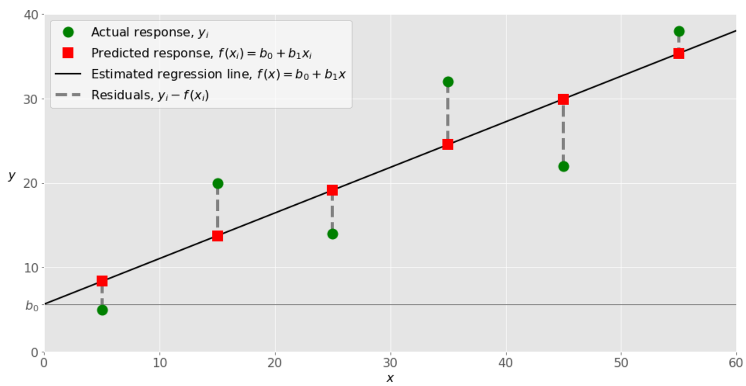
Simple Linear Regression

Simple or single-variate linear regression is the simplest case of linear regression with a single independent variable, $\mathbf{x} = x$.

The following figure illustrates simple linear regression:

When implementing simple linear regression, you typically start with a given set of input-output (x-y) pairs (green circles). These pairs are your observations. For example, the leftmost observation (green circle) has the input $x = 5$ and the actual output (response) $y = 5$. The next one has $x = 15$ and $y = 20$, and so on.

The estimated regression function (black line) has the equation $(x) = b_0 + b_1x$. Your goal is to calculate the optimal values of the predicted weights b_0 and b_1 that minimize SSR and determine the estimated regression function. The value of b_0, also called the **intercept**, shows the point where the estimated regression line crosses the y axis. It is the value of the estimated response (x) for $x = 0$. The value of b_1 determines the **slope** of the estimated regression line.



Example of simple linear regression

The predicted responses (red squares) are the points on the regression line that correspond to the input values. For example, for the input $x = 5$, the predicted response is $(5) = 8.33$ (represented with the leftmost red square).

The residuals (vertical dashed gray lines) can be calculated as y_i - $(\mathbf{x}_i) = y_i - b_0 - b_1x_i$ for $i = 1, \ldots, n$. They are the distances between the green circles and red squares. When you implement linear regression, you are actually trying to minimize these distances and make the red squares as close to the predefined green circles as possible.

Multiple Linear Regression

Multiple or multivariate linear regression is a case of linear regression with two or more independent variables.

If there are just two independent variables, the estimated regression function is $(x_1, x_2) = b_0 + b_1x_1 + b_2x_2$. It represents a regression plane in a three-dimensional space. The goal of

	regression is to determine the values of the weights b_0, b_1, and b_2 such that this plane is as close as possible to the actual responses and yield the minimal SSR. The case of more than two independent variables is similar, but more general. The estimated regression function is $(x_1, \ldots, x_r) = b_0 + b_1x_1 + \cdots + b_rx_r$, and there are $r + 1$ weights to be determined when the number of inputs is r.
Procedure:	**Implementing Linear Regression in Python** **Python Packages for Linear Regression** The package **NumPy** is a fundamental Python scientific package that allows many high-performance operations on single- and multi-dimensional arrays. It also offers many mathematical routines. Of course, it's open source. The package **scikit-learn** is a widely used Python library for machine learning, built on top of NumPy and some other packages. It provides the means for preprocessing data, reducing dimensionality, implementing regression, classification, clustering, and more. Like NumPy, scikit-learn is also open source. **Simple Linear Regression With scikit-learn** There are five basic steps when you're implementing linear regression: 1. **Import the packages and classes you need.** • Numpy • Pandas • Matplotlib • scikit-learn 2. Do the Data Preprocessing using Data Preprocessing Tools 3. Create a regression model and fit it with existing data. 4. Check the results of model fitting to know whether the model is satisfactory. 5. Apply the model for predictions. These steps are more or less general for most of the regression approaches and implementations.
Task to Perform:	1. Implement Linear Regression on dataset provided and find its accuracy 2. Implement Multiple linear regression and find its Coefficients and R Square metric
Conclusion	In this way we successfully implement Simple and Multiple linear regression and find its accuracy.

Test Your Knowledge	1. For the model you created in lab write the observed result in following Table

	Simple Linear Regression		**Multiple Linear Regression**	
Accuracy Score	Train		Train	
	Test		Test	
R2 Score				
Coefficients of model (Only 23 Decimal Points)				
Intercept of model				
Test Example Result	Predict Salary for 6.1 Year Experience =		Predict profit for { R&D Spend = 300000, Administration Spend = 160000, Marketing Spend = 200000, State = Florida}	

2. Now change the random_state parameter to any number while splitting data into train and test and write the following observations

	Simple Linear Regression		**Multiple Linear Regression**	
Accuracy Score	Train		Train	
	Test		Test	
R2 Score				
Coefficients of model (Only 23 Decimal Points)				
Intercept of model				
Test Example Result	Predict Salary for 6.1 Year Experience =		Predict profit for { R&D Spend = 300000, Administration Spend = 160000, Marketing Spend = 200000, State = Florida}	